# YEDITEPE UNIVERSITY



# MTH424 GENERATIVE AI MODELS PROJECT


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# Dataset

We used the AG News dataset, which contains **127,600 news article samples** collected from over 2,000 sources by the academic search engine ComeToMyHead, active since July 2004. The dataset is already split into **120,000 training samples** and **7,600 test samples**. Each entry consists of a short news article (text) and its corresponding class label (label). We checked the dataset for missing values and confirmed that there are no missing texts or labels.

We also analyzed the **text length distribution**, finding that the average article is **236 characters long**, with **75% of the texts under 266 characters**. This shows that the full text of each article can be used directly in both classical machine learning models and LLM-based approaches without concerns about truncation or excessive noise.

The dataset includes **four categories**:

- 0 → World
- 1 → Sports
- 2 → Business
- 3 → Sci/Tech

| | text | label |
|---|---|---|
| 0 | Wall St. Bears Claw Back Into the Black (Reute... | 2 |
| 1 | Carlyle Looks Toward Commercial Aerospace (Reu... | 2 |
| 2 | Oil and Economy Cloud Stocks' Outlook (Reuters... | 2 |
| 3 | Iraq Halts Oil Exports from Main Southern Pipe... | 2 |
| 4 | Oil prices soar to all-time record, posing new... | 2 |

**Figure 1.1** Dataset first five rows

# Dataset Split

From the default split, we used 120,000 samples for training and 7,600 samples for testing, with class distributions balanced as shown in Figure 1.2. To create a validation set, we further split the training data: 90% (108,000 samples) was saved as train_split.csv, and 10% (12,000 samples) as validation.csv, as illustrated in Figure 1.3.

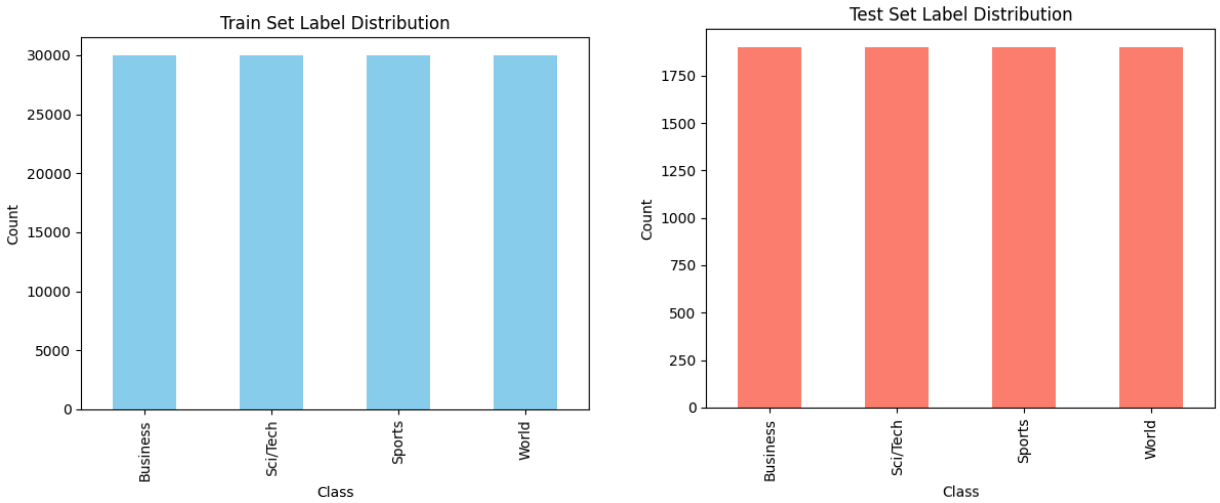



**Figure 1.2** default distribution of the labels from HuggingFace

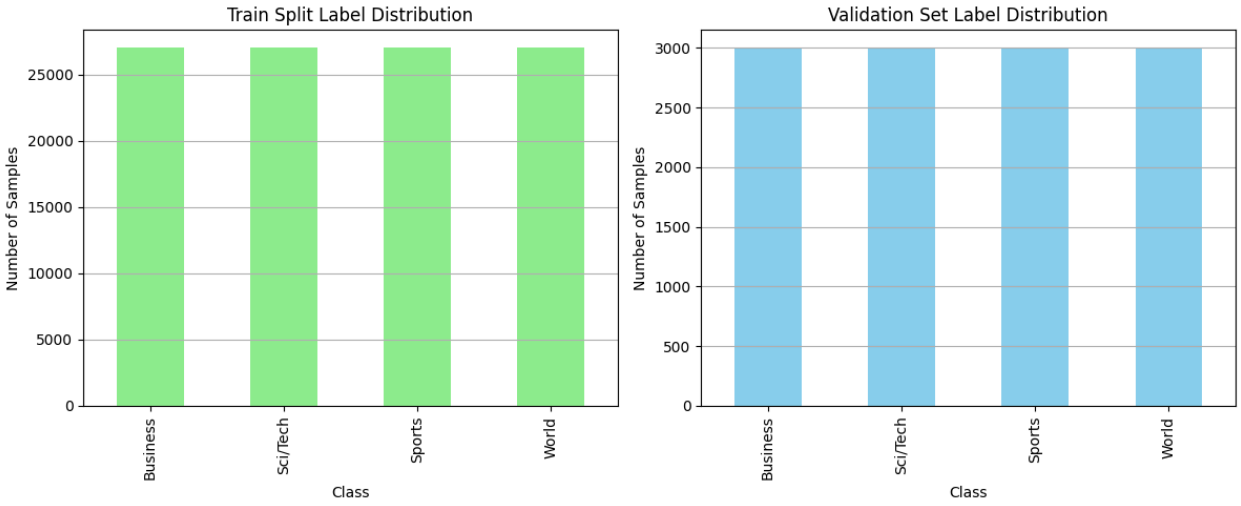



**Figure 1.3** Validation and new train set (train_split.csv) label distribution

To speed up development and experiment iterations, we created smaller subsample sets from both train_split and validation. Specifically, we randomly selected 16,000 samples from the training split and 4,000 samples from the validation set, as shown in Figure 1.4. These subsamples allowed us to rapidly test and tune models, while saving the full sets for final evaluation alongside the test set.

However, we observed that using the smaller subsample sets did not significantly reduce training time or complexity for the TF-IDF + Logistic Regression model. This is likely because TF-IDF produces sparse and efficiently processed vectors, and Logistic Regression is already a computationally lightweight model. As a result, training on the full train_split (108,000 samples) remained fast and feasible, which confirms the scalability of this classical approach.

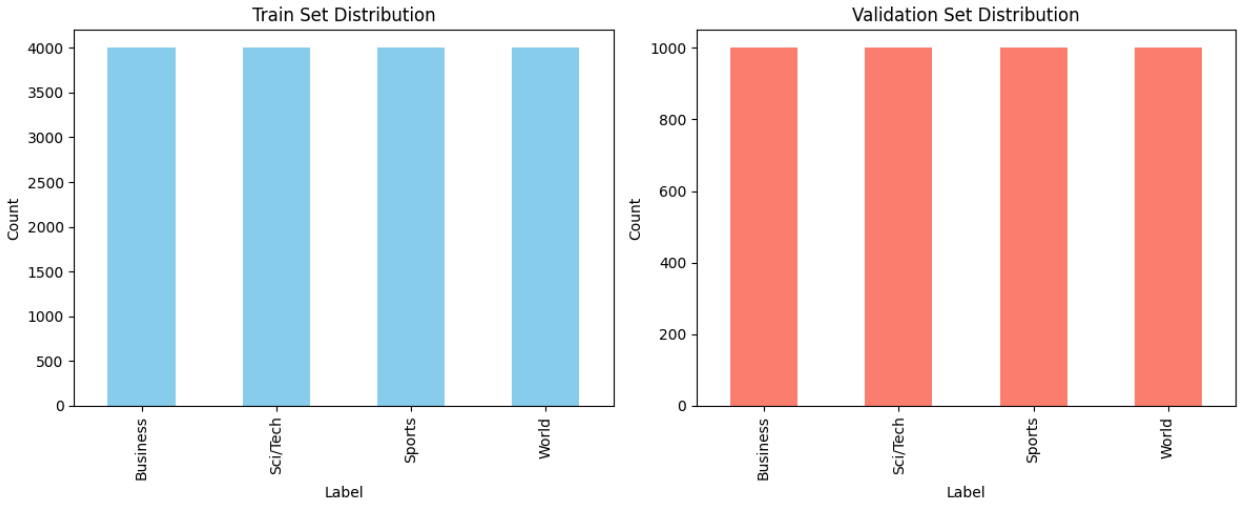



**Figure 1.4** Subsample Label Distribution

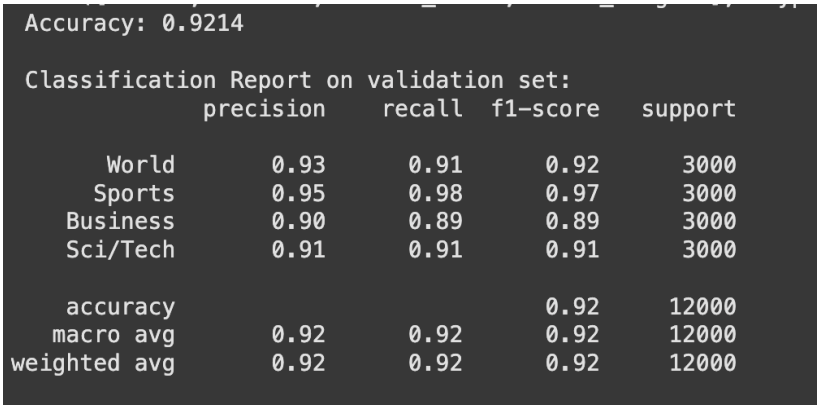
# TF-IDF and Logistic Regression

**TF-IDF (Term Frequency–Inverse Document Frequency)** is a way to turn text into numbers by measuring how important each word is in a document. TF (Term Frequency) counts how often a word appears in a specific document — more frequent words get higher scores. IDF (Inverse Document Frequency) checks how rare that word is across all documents — words that appear in many documents (like “the” or “and”) get lower scores, while rare and more informative words get higher scores. TF-IDF combines these two values to highlight the words that are both frequent in one document and rare in others. This is better than using Bag of Words, which only counts word frequency and treats all words equally, even common ones that don't carry meaning. TF-IDF helps Logistic Regression by giving it cleaner, more meaningful features — so it can better learn which words actually help distinguish between classes like "Sports" or "Business."

**Logistic Regression** is a linear classification algorithm that models the probability of each class using a weighted combination of input features. When combined with TF-IDF, which converts text into meaningful numerical vectors, Logistic Regression learns which words are most indicative of each class (e.g., "goal" → Sports, "stock" → Business). The model outputs probabilities for each class and chooses the one with the highest. It is efficient, interpretable, and performs well on linearly separable datasets like AG News.

# Subsample and Original Set's Tests

Next, we trained on the full training and validation sets (train_split.csv and validation.csv) and observed results for several max_features values.The max_features parameter limits the number of words (features) used by the model to the top N terms based on their TF-IDF scores. For example, setting max_features=5,000 keeps only the 5,000 most informative words, helping the model concentrate on meaningful patterns while ignoring overly frequent or rare terms. A value that is too low may result in lost information, while a value that is too high can introduce noise or increase training time. Therefore, tuning this parameter involves a trade-off between performance, accuracy, and computational efficiency. We found that 40,000 gave the best performance overall, as illustrated in Figure 2.3. However, we also noticed that increasing max_features beyond a certain point did not always improve accuracy — in some cases, it led to performance drops. This confirms that selecting the right vocabulary size is crucial and must be balanced carefully.



```
Accuracy: 0.9214

Classification Report on validation set:
              precision    recall  f1-score   support

       World       0.93      0.91      0.92      3000
      Sports       0.95      0.98      0.97      3000
    Business       0.90      0.89      0.89      3000
    Sci/Tech       0.91      0.91      0.91      3000

    accuracy                           0.92     12000
   macro avg       0.92      0.92      0.92     12000
weighted avg       0.92      0.92      0.92     12000
```

**Figure 2.3** train_split.csv and validation.csv results in logistic regression with 40,000 max_features

Regarding preprocessing, it's important to note that TfidfVectorizer by default only considers tokens that are two or more characters long and consist of letters, numbers, or underscores. It automatically ignores single-character tokens, punctuation, and symbols such as $, %, !, or –. This already reduces some noise. However, in our custom preprocessing step, we used a regular expression (re.sub(r"[^a-zA-Z\s]", "", text)) that removed all special characters, digits, and punctuation. While this simplified the text, it also unintentionally removed some informative tokens such as "Q2", "3-0", "$", "NASA", or "COVID-19" — which could be highly relevant for distinguishing news categories like *Business*, *Sports*, or *Sci/Tech*. As a result, we observed a slight accuracy drop from 92.14% to 91.60%. This shows that aggressive cleaning on an already clean dataset can remove useful signals and negatively affect model performance. Based on this finding, we decided not to apply custom text cleaning before training the Logistic Regression model.

Lastly, we evaluated the final Logistic Regression model using the default test set, and the results are presented in Figure 2.4. The accuracy on the test set was 91.67%, which is slightly lower than the validation set accuracy of 92.14%. This small difference is expected, as the test set originates from a separate split provided by the dataset, which may contain different vocabulary or phrasing not seen during training. In contrast, the validation set was derived from the training data and therefore shares similar distributional properties. This slight drop indicates that the model generalizes well, with no signs of overfitting.

```
Accuracy on Test Set: 0.9167

Classification Report on Test Set:
              precision    recall  f1-score   support

       World       0.93      0.90      0.92      1900
      Sports       0.96      0.98      0.97      1900
    Business       0.89      0.88      0.88      1900
    Sci/Tech       0.89      0.90      0.89      1900

    accuracy                           0.92      7600
   macro avg       0.92      0.92      0.92      7600
weighted avg       0.92      0.92      0.92      7600
```

**Figure 2.4** Logistic regression Test.csv set evaluation

# Qwen Large Language Model

This study employs the **Qwen 1.5B-Chat** model for text classification tasks. Qwen is a Transformer-based large language model developed by Alibaba, containing 500 million parameters. The Transformer architecture utilizes attention mechanisms to simultaneously process all words in a text, enabling better understanding of context and relationships between words. The chat-tuned version is specifically optimized for conversation and instruction-following tasks.

## Classification Approach

Qwen operates as an **autoregressive language model**, which differs from traditional classification models that perform direct category predictions. Instead, it generates text by continuing from given prompts. The classification approach implemented in this work leverages the model's text generation capability to produce category names.

The model tokenizes the input text and employs an **attention mechanism** to compute relationships between tokens. Subsequently, it performs **next token prediction** over the vocabulary to select the most probable next token. For example, when processing the input

"The stock market reached new highs today," the model follows system prompt instructions and generates the word "Business" through token-by-token prediction.

## Probability-Based Decision Mechanism

Unlike binary (0/1) classification systems, Qwen calculates probability distributions for each token. This implementation utilizes a **greedy decoding** strategy, selecting the token with the highest probability at each step. For instance:

- "Business": 0.82 probability
- "Sports": 0.10 probability
- "World": 0.05 probability

The model selects the "Business" token based on the highest probability score. The *max_new_tokens=20* parameter was configured to ensure concise responses.

## Chat Template and Instruction Following

Instructions were provided to the model using Qwen's specialized chat format. The *tokenizer.apply_chat_template()* function converts the system prompt and user input into a format compatible with the model's architecture. Through this format, the model interprets the system prompt as role instructions and processes the user text as a classification task.

## Output Processing and Error Handling

Post-processing techniques were applied to convert Qwen's text output into numeric labels. Since the model occasionally generates additional words such as "Business news," output parsing was implemented to extract only category names. Invalid formats were labeled as -1 and filtered out, with performance metrics calculated exclusively on valid predictions.

This methodology differs from classical machine learning approaches by eliminating the need for model training and achieving effective classification solely through prompt engineering. However, this approach results in lower inference speed and requires additional processing to control output format.

# Evaluation and Improvement of Results

Accuracy, F1-score, precision, and recall are the primary metrics for evaluating model performance. Initially, the dataset was analyzed to determine a reasonable expected outcome. A baseline was then established using TF-IDF and logistic regression. Then, the model was tested with a simple default (zero-shot) prompt to get its initial performance on this dataset.

After calculating the evaluation metrics and their averages, the prompt was improved through experimental iterations. The modifications and their effects on performance are detailed below.

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**Attempt 1: Default (Zero-Shot) Prompt**

Only the expected output is explained in the prompt for the baseline.

*"You are a news article classifier. You must classify news headlines into one of these categories: [World, Sports, Business, Sci/Tech]. Only respond with the category name."*

📊 Classification Report:

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| World | 0.97 | 0.04 | 0.07 | 996 |
| Sports | 0.72 | 0.84 | 0.78 | 1000 |
| Business | 0.35 | 0.96 | 0.51 | 996 |
| Sci/Tech | 0.89 | 0.02 | 0.05 | 984 |
| accuracy | | | 0.47 | 3976 |
| macro avg | 0.73 | 0.47 | 0.35 | 3976 |
| weighted avg | 0.73 | 0.47 | 0.35 | 3976 |

*World and Sci/Tech F1-Scores are unexpectedly low.*

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**Attempt 2: More Detailed Prompt**

Before trying few-shot, more explanation about the labels given.

*"You are a news article classifier. You must classify news headlines into one of these categories: [World, Sports, Business, Sci/Tech]. Only respond with the category name.*
***Here is some information about the topics that may help:***
*1. **World** — News about international politics, government policies, diplomacy, law, conflicts, elections, or global social issues.*
*2. **Business** — News about companies, stock markets, investments, profits, economic trends, industries, or corporate strategy.*
*3. **Science/Technology** — News about new technology, inventions, software, hardware, science discoveries, space, or research.*
*4. **Sports** — News about sports teams, athletes, competitions, scores, matches, seasons, and sporting events."*

📊 Classification Report:

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| World | 0.76 | 0.10 | 0.17 | 957 |
| Sports | 0.56 | 0.77 | 0.65 | 993 |
| Business | 0.37 | 0.88 | 0.52 | 982 |
| Sci/Tech | 0.86 | 0.07 | 0.13 | 957 |
| accuracy | | | 0.46 | 3889 |
| macro avg | 0.64 | 0.45 | 0.37 | 3889 |
| weighted avg | 0.64 | 0.46 | 0.37 | 3889 |

*This approach improved the World and the Sci/Tech scores, but didn't make a significant change.*

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**Attempt 3: Few-Shot Prompt**

Articles with incorrect predictions were examined for each label. The most commonly misclassified words for each label were provided in the prompt using few-shot examples.

*"You are a news article classifier. You must classify news headlines into one of these categories: [World, Sports, Business, Sci/Tech]. Only respond with the category name.*
*- World articles **often include terms like**: UN, Palestinian, Gaza, peace, election, Israel, minister, nuclear, president, Iraq, Iran, Afghanistan, rebel, asylum, intervention, Putin, Trump, veto.*
*- Sports articles often include terms like: champion, score, cup, baseball, Olympic, win, sports, football, NHL, locker room, Formula One, Super Bowl, cricket. Anything Sports Network news agency reports should be labeled as Sports. Also if it has a branch of sports or a sports team name it is highly a sports article.*
*- Business articles often include terms like: stock, supply, profit, economic, airlines, company, bank, inc, dollar, tax, billion, finance, economy, gas, oil, union, currency, export, merger, sales, market, trade, revenue, shareholder.*
*- Sci/Tech articles often include terms like: Apple, music, technology, computer, Microsoft, software, scientist, Nasa, space, internet, engineering, gamer, Intel, theory, satellite, programming. If there is a game name like Scribe, Grand Theft Auto it is highly a sci/tech article. Also if it's an environmental issue or an animal species issue it may belong to Sci/Tech."*

📊 Classification Report:

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| World | 0.89 | 0.03 | 0.06 | 974 |
| Sports | 0.58 | 0.89 | 0.70 | 994 |
| Business | 0.42 | 0.72 | 0.53 | 976 |
| Sci/Tech | 0.44 | 0.32 | 0.37 | 974 |
| accuracy | | | 0.49 | 3918 |
| macro avg | 0.58 | 0.49 | 0.42 | 3918 |
| weighted avg | 0.58 | 0.49 | 0.42 | 3918 |

*Few-shot improved the accuracy as expected.*

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**Attempt 4: Experimental Prompt**

While examining the dataset for few-shot examples, it was observed that some news items that logically should not fall under Sci/Tech (but also do not belong to the other three labels) were still categorized as Sci/Tech. Additionally, the model struggled with ambiguous news items that are between two categories and could not be distinctly classified. It was observed that few-shot examples helped improve performance for the categories where such confusion was more frequent, like Business and Sci/Tech, whereas the opposite effect was seen in World and Sports. As an experiment, few-shot prompting was applied to only half of the labels. This resulted in the worst performance overall.

*"You are a news article classifier. You must classify news headlines into one of these categories: [World, Sports, Business, Sci/Tech]. Only respond with the category name.*
*- **Business** articles often include terms like: stock, supply, profit, economic, airlines, company, bank, inc, dollar, tax, billion, finance, economy, gas, oil, union, currency, export, merger, sales, market, trade, revenue, shareholder.*
*- **Sci/Tech** articles often include terms like: Apple, music, technology, computer, Microsoft, software, scientist, Nasa, space, internet, engineering, gamer, Intel, theory, satellite, programming. If there is a game name like Scribe, Grand Theft Auto it is highly a sci/tech article. Also if it's an environmental issue or an animal species issue it may belong to Sci/Tech."*

📊 Classification Report:

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| World | 1.00 | 0.00 | 0.00 | 995 |
| Sports | 0.90 | 0.51 | 0.65 | 999 |
| Business | 0.31 | 0.94 | 0.46 | 993 |
| Sci/Tech | 0.44 | 0.17 | 0.24 | 997 |
| accuracy | | | 0.40 | 3984 |
| macro avg | 0.66 | 0.40 | 0.34 | 3984 |
| weighted avg | 0.66 | 0.40 | 0.34 | 3984 |

*This approach brought the lowest accuracy.*

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**Attempt 5: 2. Experimental Prompt (FINAL)**

Few-shot examples are given for all the labels. But few-shots of labels with low F1-scores were simplified. This prompt is chosen as the **final** because of the **highest accuracy** rate.

*"You are a news article classifier. You must classify news headlines into one of these categories: [World, Sports, Business, Sci/Tech]. Only respond with the category name.*
*- World articles often include terms like: UN, Palestinian, Gaza, peace, election, Israel, minister, president, Iraq, Iran, Afghanistan, Putin, Trump. News about international politics, government policies, diplomacy, law, conflicts, elections, or global social issues.*

*- Sports articles often include terms like: champion, score, cup, baseball, Olympic, sports, football, NHL, cricket, Sports Network.*
*- Business articles often include terms like: stock, supply, profit, economic, bank, inc, dollar, tax, finance, economy, gas, oil, union, currency, export, sales, market, trade, revenue.*
*- Sci/Tech articles often include terms like: Apple, technology, computer, Microsoft, software, scientist, Nasa, space, internet, engineering, gamer, Intel, theory, satellite, programming. If there is a game name like Scribe, Grand Theft Auto it is highly a sci/tech article. Also if it's an environmental issue or an animal species issue it may belong to Sci/Tech."*

📊 Classification Report:

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| World | 0.88 | 0.06 | 0.11 | 976 |
| Sports | 0.65 | 0.89 | 0.75 | 998 |
| Business | 0.44 | 0.69 | 0.54 | 985 |
| Sci/Tech | 0.40 | 0.40 | 0.40 | 980 |
| accuracy | | | 0.51 | 3939 |
| macro avg | 0.60 | 0.51 | 0.45 | 3939 |
| weighted avg | 0.59 | 0.51 | 0.45 | 3939 |

*The **best accuracy** without fine-tuning achieved.*

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**Attempt 5: Improving F1-Score Prompt**

It was observed that the recall value for data labeled as World was very low. Since the model often confused World news with Business news, it was considered that this issue could be addressed by lowering the threshold for selecting the World label. Prompt-based fix was not effective; better results might be achieved by adjusting the model's weights.

*"You are a news article classifier. You must classify news headlines into one of these categories: [World, Sports, Business, Sci/Tech]. Only respond with the category name.* ***Keep the treshold for World articles low because the precision is way higher than recall for World.***
*- World articles often include terms like: UN, Palestinian, Gaza, peace, election, Israel, minister, president, Iraq, Iran, Afghanistan, Putin, Trump. News about international politics, government policies, diplomacy, law, conflicts, elections, or global social issues.*
*- Sports articles often include terms like: champion, score, cup, baseball, Olympic, sports, football, NHL, cricket, Sports Network.*
*- Business articles often include terms like: stock, supply, profit, economic, bank, inc, dollar, tax, finance, economy, gas, oil, union, currency, export, sales, market, trade, revenue.*
*- Sci/Tech articles often include terms like: Apple, technology, computer, Microsoft, software, scientist, Nasa, space, internet, engineering, gamer, Intel, theory, satellite, programming. If there is a game name like Scribe, Grand Theft Auto it is highly a sci/tech article. Also if it's an environmental issue or an animal species issue it may belong to Sci/Tech."*

📊 Classification Report:

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| World | 0.87 | 0.03 | 0.05 | 971 |
| Sports | 0.64 | 0.87 | 0.74 | 1000 |
| Business | 0.40 | 0.78 | 0.53 | 976 |
| Sci/Tech | 0.43 | 0.27 | 0.33 | 985 |
| accuracy | | | 0.49 | 3932 |
| macro avg | 0.59 | 0.49 | 0.41 | 3932 |
| weighted avg | 0.59 | 0.49 | 0.42 | 3932 |

*Low recall problem couldn't be solved by the prompt. Weight adjustments may be needed.*

---

In conclusion, several factors are believed to contribute to the model's low accuracy:

- Ambiguities in dataset labeling (overlap between World and Business news, unrelated Sci/Tech labels, and the need for domain knowledge or player names to interpret Sports news),
- An inefficient threshold for classifying World news,
- General model limitations (other LLMs demonstrate higher accuracy on similar tasks according to Hugging Face benchmarks),
- The model's training date (Qwen struggled with recent topics such as Israel's genocide in Gaza, as our dataset primarily consists of news from the past year).

Classification Report:

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| World | 0.88 | 0.08 | 0.14 | 969 |
| Sports | 0.65 | 0.89 | 0.76 | 1000 |
| Business | 0.42 | 0.66 | 0.51 | 982 |
| Sci/Tech | 0.41 | 0.39 | 0.40 | 979 |
| accuracy | | | 0.51 | 3930 |
| macro avg | 0.59 | 0.51 | 0.45 | 3930 |
| weighted avg | 0.59 | 0.51 | 0.45 | 3930 |

*Validation Results with Attempt 5 prompt*

Classification Report:

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| World | 0.84 | 0.07 | 0.13 | 1845 |
| Sports | 0.64 | 0.89 | 0.75 | 1899 |
| Business | 0.43 | 0.66 | 0.52 | 1880 |
| Sci/Tech | 0.40 | 0.39 | 0.40 | 1871 |
| accuracy | | | 0.51 | 7495 |
| macro avg | 0.58 | 0.50 | 0.45 | 7495 |
| weighted avg | 0.58 | 0.51 | 0.45 | 7495 |

*Test Results with Attempt 5 prompt*

# Fine-Tuning Qwen 1.5b

Since prompt-based approaches did not yield the expected improvements in Qwen's classification accuracy, we turned to fine-tuning as a more effective solution. Fine-tuning a large language model (LLM) allows for customizing its behavior, injecting domain-specific knowledge, and optimizing performance for specialized tasks. By training a pretrained model on a targeted dataset, we can:

- **Incorporate New Knowledge:** Introduce domain-relevant concepts and context that may not be well represented in the base model.
- **Adapt Model Behavior:** Tailor the model's tone, response structure, or style to align with the requirements of the task or domain.
- **Enhance Task Performance:** Improve classification accuracy, consistency, and relevance for the specific use case.

To perform this fine-tuning efficiently, we used LoRA (Low-Rank Adaptation), a parameter-efficient method that reduces memory and processing costs, while not compromising performance. LoRA adds small trainable low-rank matrices to specific layers of the model while keeping the original weights frozen. This approach significantly reduces memory and computational demands, making fine-tuning feasible on modest hardware without sacrificing performance.

At the beginning of the code, essential libraries are loaded: transformers for model and tokenizer operations, peft for applying LoRA, accelerate for training optimization and memory management, and datasets for loading and processing datasets. The AG News dataset, which

is well-suited for text classification tasks, is then loaded and downsampled to smaller subsets for rapid experimentation.

The core of the fine-tuning process lies in the configuration of LoRA. Using the LoraConfig class, key hyperparameters such as the rank (r), scaling factor (lora_alpha), and dropout rate (lora_dropout) are defined. Of particular importance is the target_modules parameter, which specifies the layers of the model where LoRA adapters will be injected. In our case, the query (q_proj) and value (v_proj) projection layers within the attention mechanism were selected as targets.

The original Qwen model is loaded using AutoModelForCausalLM.from_pretrained, and the LoRA configuration is applied through the get_peft_model function. This process freezes the original model weights and adds a small set of trainable LoRA adapters, allowing the model to be fine-tuned efficiently without modifying its full parameter set.

During the preprocessing stage, each headline and its associated label are converted into numerical token IDs using the AutoTokenizer. A custom function named tokenize_for_classification_ft formats each example as a prompt-response pair, combining the instruction and label into a single input. The sequences are padded to a fixed maximum length, and attention masks are generated to help the model focus only on meaningful tokens while ignoring padding tokens.

The training setup is defined using the TrainingArguments class, where various hyperparameters such as output directory, batch size, learning rate, and number of epochs are configured. Then, the Trainer class integrates the model, training arguments, training and validation datasets, and a data collator to manage the training loop. With the trainer.train() command, fine-tuning is initiated, and the model is trained on the target dataset over the specified number of epochs.

Once training is complete, the fine-tuned model weights and tokenizer configuration are saved. In the evaluation phase, the model's performance on the test dataset is assessed by generating predictions using a text generation pipeline and comparing those predictions with the ground-truth labels. Helper functions such as normalize_label are used to standardize the model's outputs to match the expected label format.

After the fine-tuning process, the overall accuracy can be evaluated as 71%. However, when the performance is examined on a class basis, it is seen that the model performs quite well in some classes (such as "sports"), while it has more difficulty in others (such as "sci/tech" and "world"). The high performance in the "Sports" class shows that the model is quite successful in distinguishing news headlines in this category. The low performance in the "Sci/Tech" class may indicate that this class may contain more complex or ambiguous language than other classes, which may make it difficult for the model to classify correctly. In addition, the model shows that almost all of the news it labels as "world" really belong to this category. However, it may be

missing a significant portion of the news in the real "world" category, i.e., it may be assigning them to other categories. In the fine-tuning process, performance can be increased by trying different values of fine-tuning hyperparameters such as learning rate and epoch number.

```
Classification Report:

              precision    recall  f1-score   support

    business      0.686     0.750     0.716        32
    sci/tech      0.385     0.526     0.444        19
      sports      0.931     1.000     0.964        27
       world      1.000     0.455     0.625        22

    accuracy                         0.710       100
   macro avg      0.750     0.683     0.688       100
weighted avg      0.764     0.710     0.712       100
```

Classification report on all test data (test.csv):

```
Classification Report:

              precision    recall  f1-score   support

    business      0.647     0.735     0.688      1900
    sci/tech      0.536     0.661     0.592      1896
      sports      0.932     0.986     0.958      1899
       world      0.970     0.554     0.705      1897

    accuracy                         0.734      7592
   macro avg      0.771     0.734     0.736      7592
weighted avg      0.771     0.734     0.736      7592
```

**Hugging Face repo URL:** https://huggingface.co/ecettt/qwen-1.5b-finetuned

# Conclusion

This study conducted a systematic evaluation of classical machine learning versus modern large language model approaches for news text classification using the AG News dataset. The comparative analysis reveals significant performance differences that provide important insights for methodology selection in structured text classification tasks.

The TF-IDF + Logistic Regression baseline demonstrated superior performance across all evaluation metrics, achieving 92.14% validation accuracy and 91.67% test accuracy. This classical approach exhibited balanced performance with particularly strong results in Sports (F1: 0.97) and robust classification across World, Business, and Sci/Tech categories. In contrast, prompt-based approaches with Qwen 1.5B-Chat achieved substantially lower performance despite systematic prompt engineering efforts, with the zero-shot baseline yielding 47% accuracy and the most optimized few-shot prompt achieving 51% accuracy. These results represent a significant 41-point performance gap compared to the classical baseline.

LoRA fine-tuning of Qwen 1.5B demonstrated the value of supervised adaptation, improving accuracy from 51% to 73%, yet still underperforming the classical approach by over 20 percentage points. The fine-tuning results revealed category-specific patterns, with strong Sports classification (F1: 0.958) but continued challenges in World (F1: 0.705) and Sci/Tech (F1: 0.592) categories, indicating persistent architectural limitations for fine-grained text classification.

Since prompt-based classification did not yield sufficient accuracy, the Qwen 1.5B model is fine-tuned using LoRA. This supervised adaptation significantly improved performance, raising test accuracy from 51% to 73%. However, it still lagged behind the classical tf-idf + logistic regression baseline by over 20 percentage points. Fine-tuning revealed category-specific strengths and weaknesses: the model excelled in classifying Sports (F1: 0.958), but struggled with World (F1: 0.705) and Sci/Tech (F1: 0.444). These results underscore the limitations of the model’s architecture in handling fine-grained distinctions, despite gains from task-specific training.

Several important insights emerged from this analysis. The AG News dataset's characteristics—short headlines, balanced class distribution, and clear keyword-based distinctions—favor classical approaches that excel at pattern recognition in structured text. Our systematic prompt engineering experiments demonstrated that architectural limitations cannot be overcome through prompt optimization alone, while aggressive text preprocessing proved counterproductive by removing informative tokens that carry strong categorical signals. The classical approach offers substantial advantages in computational efficiency, interpretability, and consistent performance across categories, providing optimal accuracy with minimal resource requirements.

This research demonstrates that methodology selection should be guided by task characteristics rather than technological novelty. For well-defined text classification problems

with clear linguistic patterns, classical machine learning methods remain highly competitive and often superior to modern large language models. The substantial performance gaps observed emphasize the importance of empirical evaluation in determining optimal approaches for specific application domains.